

## NABOUWALU, Fiji

WMO: 916590

Lat: 17.000S

Lon: 178.700E

Elev: 36

StdP: 100.89

Time Zone: 12.00 (PCF)

Period: 82-10

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 20.9       | 21.5 | 14.2                          | 10.1 | 23.1 | 15.2 | 10.8 | 23.3 | 14.9                  | 22.7 | 12.7 | 23.6 | 4.2                   | 120  | 0.388 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 4.6                    | 31.2            | 26.6 | 30.7 | 26.3 | 30.2 | 26.0 | 27.3                | 30.2 | 27.0 | 29.9 | 26.6 | 29.4 | 2.9                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.5                            | 22.1 | 29.6 | 26.1 | 21.6 | 29.2 | 25.7 | 21.1 | 28.7 | 87.1          | 30.5 | 85.2 | 30.0 | 83.5 | 29.6 | 29.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |      |     |                    | n-Year Return Period Values of Extreme Temperature |           |     |            |     |            |     |            |     |
|-------------------|-----|------|----------------------------|------|-----|--------------------|----------------------------------------------------|-----------|-----|------------|-----|------------|-----|------------|-----|
| 1%                |     | 2.5% | 5%                         | Mean |     | Standard Deviation |                                                    | n=5 years |     | n=10 years |     | n=20 years |     | n=50 years |     |
| Min               |     | Max  |                            | Min  | Max |                    | Min                                                | Max       | Min | Max        | Min | Max        | Min | Max        |     |
| (a)               | (b) | (c)  |                            | (e)  | (f) | (g)                | (h)                                                | (i)       | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 10.3              | 9.4 | 8.6  |                            | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |     |      | DB                         |      |     |                    |                                                    |           |     |            |     |            |     |            |     |
|                   |     |      | WB                         |      |     |                    |                                                    |           |     |            |     |            |     |            |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.9   | 27.3  | 27.4  | 27.3  | 26.8  | 25.7  | 25.1  | 24.3  | 24.3  | 24.8  | 25.2  | 26.0  | 26.7 |
|                                                                        | DBStd       | 1.53   | 0.97  | 0.92  | 0.90  | 1.03  | 0.97  | 0.99  | 1.06  | 1.21  | 1.20  | 1.16  | 1.12  | 1.02 |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                        | CDD10.0     | 5805   | 536   | 486   | 536   | 505   | 487   | 452   | 443   | 444   | 444   | 473   | 481   | 518  |
|                                                                        | CDD18.3     | 2763   | 277   | 252   | 278   | 255   | 229   | 202   | 185   | 186   | 194   | 214   | 231   | 259  |
|                                                                        | CDH23.3     | 21287  | 2748  | 2528  | 2756  | 2335  | 1661  | 1184  | 759   | 826   | 1046  | 1312  | 1777  | 2356 |
| (14) Wind                                                              | WSAvg       | 3.9    | 3.1   | 2.7   | 2.6   | 3.1   | 3.7   | 4.2   | 4.7   | 4.7   | 5.0   | 4.9   | 4.0   | 3.6  |
|                                                                        | PrecAvg     | 2313   | 354   | 282   | 290   | 247   | 175   | 106   | 89    | 110   | 106   | 160   | 185   | 247  |
|                                                                        | PrecMax     | 3571   | 1280  | 558   | 548   | 710   | 417   | 265   | 222   | 346   | 392   | 433   | 554   | 422  |
|                                                                        | PrecMin     | 1352   | 97    | 93    | 101   | 66    | 44    | 19    | 14    | 13    | 11    | 8     | 44    | 71   |
|                                                                        | PrecStd     | 519    | 230   | 112   | 118   | 141   | 99    | 62    | 50    | 79    | 76    | 123   | 120   | 97   |
|                                                                        | DB          | 31.7   | 31.6  | 31.8  | 31.4  | 30.2  | 28.9  | 28.7  | 28.5  | 29.1  | 29.7  | 30.7  | 31.1  |      |
|                                                                        | MCWB        | 26.9   | 26.9  | 27.2  | 26.7  | 25.7  | 24.7  | 24.8  | 23.8  | 25.3  | 24.9  | 26.0  | 26.4  |      |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 30.9   | 30.9  | 30.9  | 30.3  | 29.0  | 27.9  | 27.5  | 27.5  | 28.2  | 28.8  | 29.7  | 30.3  |      |
|                                                                        | 2% MCWB     | 26.4   | 26.6  | 26.6  | 26.2  | 25.1  | 24.3  | 23.7  | 23.6  | 24.4  | 24.7  | 25.4  | 26.0  |      |
|                                                                        | 5% DB       | 30.3   | 30.4  | 30.2  | 29.6  | 28.2  | 27.3  | 26.7  | 26.8  | 27.5  | 28.1  | 29.0  | 29.6  |      |
|                                                                        | 5% MCWB     | 26.1   | 26.3  | 26.2  | 25.8  | 24.4  | 23.8  | 23.2  | 23.3  | 23.9  | 24.2  | 25.0  | 25.6  |      |
|                                                                        | 10% DB      | 29.7   | 29.7  | 29.5  | 28.9  | 27.5  | 26.7  | 26.0  | 26.2  | 26.8  | 27.3  | 28.3  | 28.9  |      |
|                                                                        | 10% MCWB    | 25.7   | 25.9  | 25.9  | 25.4  | 24.0  | 23.5  | 22.6  | 22.8  | 23.5  | 23.6  | 24.6  | 25.1  |      |
|                                                                        | WB          | 27.7   | 27.9  | 27.9  | 27.4  | 26.2  | 25.5  | 25.5  | 25.0  | 25.7  | 26.2  | 27.0  | 27.2  |      |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% MCDB   | 30.6   | 30.5  | 30.9  | 30.4  | 29.0  | 27.7  | 28.0  | 27.1  | 28.3  | 29.1  | 30.1  | 30.1  |      |
|                                                                        | 2% WB       | 27.1   | 27.2  | 27.2  | 26.7  | 25.7  | 25.0  | 24.6  | 24.4  | 25.0  | 25.4  | 26.1  | 26.6  |      |
|                                                                        | 2% MCDB     | 30.1   | 30.0  | 30.1  | 29.6  | 28.3  | 27.2  | 26.7  | 26.6  | 27.5  | 28.0  | 28.9  | 29.5  |      |
|                                                                        | 5% WB       | 26.6   | 26.8  | 26.7  | 26.2  | 25.2  | 24.6  | 24.1  | 24.0  | 24.5  | 24.8  | 25.6  | 26.1  |      |
|                                                                        | 5% MCDB     | 29.5   | 29.6  | 29.4  | 28.9  | 27.5  | 26.7  | 26.0  | 26.2  | 26.8  | 27.3  | 28.2  | 28.9  |      |
|                                                                        | 10% WB      | 26.1   | 26.4  | 26.2  | 25.8  | 24.7  | 24.1  | 23.6  | 23.6  | 24.0  | 24.4  | 25.1  | 25.6  |      |
|                                                                        | 10% MCDB    | 28.9   | 29.1  | 28.8  | 28.3  | 26.9  | 26.2  | 25.5  | 25.8  | 26.3  | 26.7  | 27.7  | 28.3  |      |
| (35) Mean Daily Temperature Range                                      | MDBR        | 4.7    | 4.6   | 4.6   | 4.2   | 3.5   | 3.3   | 3.3   | 3.3   | 3.5   | 3.7   | 4.1   | 4.2   |      |
|                                                                        | 5% DB MCDBR | 5.5    | 5.6   | 5.6   | 5.3   | 4.6   | 4.1   | 4.3   | 4.0   | 4.4   | 4.7   | 5.2   | 5.2   |      |
|                                                                        | 5% DB MCWBR | 2.9    | 3.0   | 3.0   | 2.8   | 2.6   | 2.5   | 2.6   | 2.1   | 2.6   | 2.6   | 2.9   | 2.8   |      |
|                                                                        | 5% WB MCDBR | 5.3    | 5.2   | 5.2   | 5.0   | 4.5   | 3.9   | 3.8   | 3.6   | 4.1   | 4.4   | 4.9   | 5.0   |      |
|                                                                        | 5% WB MCWBR | 3.1    | 3.1   | 3.1   | 2.9   | 2.7   | 2.7   | 2.7   | 2.3   | 2.7   | 2.7   | 3.1   | 3.1   |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.390  | 0.383 | 0.379 | 0.379 | 0.381 | 0.378 | 0.367 | 0.368 | 0.385 | 0.387 | 0.386 | 0.386 |      |
|                                                                        | taud        | 2.611  | 2.641 | 2.660 | 2.650 | 2.610 | 2.604 | 2.628 | 2.628 | 2.567 | 2.580 | 2.605 | 2.621 |      |
|                                                                        | Ebn at Noon | 955    | 951   | 934   | 899   | 861   | 846   | 867   | 897   | 913   | 936   | 952   | 958   |      |
|                                                                        | Edh at Noon | 103    | 99    | 94    | 90    | 88    | 85    | 85    | 90    | 102   | 104   | 103   | 102   |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 6.03   | 5.65  | 5.31  | 4.67  | 4.18  | 3.81  | 4.07  | 4.55  | 5.17  | 5.68  | 5.88  | 5.91  |      |
|                                                                        | RadStd      | 0.64   | 0.51  | 0.41  | 0.35  | 0.29  | 0.26  | 0.18  | 0.39  | 0.35  | 0.43  | 0.56  | 0.32  |      |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) | +0.18 | N/A     | N/A | N/A    | N/A     | +0.20 | +0.24 | N/A         | N/A     | +67     | +66     |

Nomenclature: See separate page